Frontier



Brown University
R&D application
2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Brown University

Organization location (we welcome applicants from anywhere in the world)

Providence RI, USA

Name(s) of primary point(s) of contact for this application

Administrative Point of Contact
Isabelle Hart
Grants and Contracts Administrator

Principal Investigator
Emily H. G. Cooperdock

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Research University

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Optimizing carbonate mineralization in ultramafic mine tailings by overcoming surface passivation

[1] We use "project" throughout this template, but the term is not intended to denote a single facility or experimental setup. The "project" being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your **core research goa**l and the **benefits you think it could provide to the broader CDR field**. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

We propose an integrated *field-based, experimental, and modeling* study to quantify, optimize, and verify CO_2 mineralization in ultramafic mine tailings for enhanced surficial mineralization applications. A major bottleneck in carbon dioxide removal is the ***limited understanding of how surface mineralization develops in natural systems***, and the subsequent impacts on the rate and efficiency of carbonation reactions in ultramafic rocks. These rates are governed by a complex interplay of factors that can either passivate or enhance carbon mineralization, including mineralogy, reactive surface area, CO_2 availability, and environmental conditions (e.g., temperature, precipitation, and soil/regolith development).

Our proposal is motivated by *Frontier's priority innovation area* to ***enhance passive surface mineralization*** at low-to-medium cost in unstructured rock piles. Surface mineralization is a pervasive feature of natural rock weathering, but the type of mineralization has large unconstrained effects on CDR potential. Ultramafic silicate minerals have a high theoretical potential to react with CO_2 to form stable carbonate minerals in minimally engineered/managed settings, however the process faces several practical challenges for commercial applications. First, slow carbonate mineralization rates require energy and additional cost to accelerate natural reactions. Commonly this involves grinding and spreading rock materials or engineering piles/reactors to increase surface area for reactions. Second, a more **problematic and underappreciated** issue is the formation of **passivating surface minerals on rock surfaces** that hinder or prohibit CDR, including amorphous Si coatings and Fe-oxides (hydroxides) which are commonly found in natural field settings. Increased surface area by grinding can exacerbate passivating mineralization, requiring further costs.

Ultramafic mine tailings in the US and Canada are a compelling resource for CDR because they are locations with already exposed reactants (ultramafic rocks and minerals) that do not require substantial acquisition of new land, increasing the likelihood of social acceptance and future government and industry investment. However, our limited understanding of the positive and negative feedbacks of passive surface mineralization in heterogeneous natural systems remains one of the greatest challenges in making CDR in unstructured rock piles economically viable on timescales necessary to impact global CO_2 budgets (10 to 50-70 years). In order to make strategic CDR investments within these industrial reservoirs, we first need to understand what surface mineral coatings are already formed, the factors that control their development, whether they are passivating (i.e., preventing further reactions), and if they are, identify potential low-cost mitigation interventions such as promotion of soil/regolith development, which can reverse the passivating role played by surface minerals.

We propose a series of “natural” and “laboratory” experiments on historic ultramafic mine tailings in CA, WA, VT, and Quebec to: 1) test the role of mineralogy (chemistry, microstructure) and soil/regolith on natural surface mineralization of unstructured rock piles over time (e.g., exposure age) and space (e.g., depth); and 2) conduct controlled laboratory experiments to evaluate how temperature, pressure, and pCO_2 can be optimized to change the outcomes of surface mineralization and enhance carbonation reaction rates as a guide for future surficial weathering in minimally managed or unmanaged (‘open system’ CDR approaches) or engineered systems (‘closed system’ CDR approaches). A novelty of our approach is that we will be able to calculate and cross-calibrate ‘natural’ and ‘laboratory’ mineralization rates over decadal timescales. An additional novelty is testing the role of soil development on mineralization and rates in mine tailing deposits.

Core questions that motivate our proposed work include: How does the mineral chemistry and microstructure of the source rock impact the capacity for carbonation reactions to take place? How are the relevant reaction rates influenced by regional climate (temperature, precipitation)? How critical is soil/regolith development for ***overcoming*** passivating surface mineralization? For example, 1) can soil products such as clay and organic matter facilitate aqueous enhancement of mineral carbonation; 2) do iron oxyhydroxides, that form protective (i.e., passivating) surfaces on ultramafic rock at the surface, facilitate soil organic matter complexation and accumulation in the sub-surface - thereby naturally flipping an agent of passivation to an agent of enhanced carbonation?

The following three breakthroughs needs to happen to unlock a step-change:

1) **The creation of a predictive framework to mitigate passivation.** A transition from site-specific trial-and-error to a predictive model of mineral coatings could be a step-change advancement in present and future applications. If we can identify the specific environmental or chemical "tipping points" that prevent amorphous silica or iron-oxide passivation, we can unlock the reactivity of millions of tons of existing tailings without the energy intensity of re-grinding.
2) **Optimization of passive (long-tail) systems.** We need to define the ideal pCO_2, moisture (pore space humidity/water content/fluid composition), and temperatures for the diversity of ultramafic lithologies to design low-cost interventions that increase CO_2 mineralization rates. If we can identify the processes and quantify the role of soils at ‘naturally’ enhancing mineralization, as well as isolating key variables in controlled lab experiments, we can prescribe recommendations based on environmental and cost factors.
3) **Verification and reliability of surficial weathering scalability and permanency.** We will quantify the positive and negative relationships between intrinsic mineralogy, CO_2 fluxes, soil/regolith development and reaction rates. Low-cost, high-confidence MRV leading to commercial-scale investment in surface mineralization projects requires this foundational data.

In addition to performing the natural and laboratory experiments outlined in this proposal, deliverables include ***developing a decision framework*** with Frontier for project developers and investors. This framework will utilize our site-specific data on mineral and soil chemistry, climate variables, and passivation thresholds to determine the key factors needed to categorize new

ultramafic sites by their 'reactivity potential.' By integrating our experimental kinetic laws with real-world field observations, this tool will help stakeholders determine where mechanical intervention (like tilling or crushing) is required and where natural biogeochemical catalysts can be leveraged (like promoting vegetation and soil development), with the goal of effectively helping to lower the barrier to entry for commercial-scale surficial mineralization projects. The project's impact extends beyond direct carbon sequestration to include environmental remediation by transforming hazardous asbestos-bearing tailings into inert minerals at legacy mining sites.

c. **Project objectives:** Describe your project in more detail: what are your **key research questions** and **proposed approach** for answering them, including **methods, project location, and scale** where applicable? Tell us why your **approach is best-in-class,** and how you're **differentiated from any other organization** working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

A central question in the efficacy of surface mineralization as a CDR technique is whether there are scalable practices that can be practically applied to existing ultramafic mine tailings to permanently store carbon dioxide as carbonate minerals in situ. Prior experimental work has focused on the conditions of surface mineralization under controlled conditions and simple mineral-fluid systems, but fewer studies have explored the same questions in natural samples where these factors are much more variable. Closing this knowledge gap is critically important for implementing, modeling and scaling CRD in natural mine tailing heaps.

Our study will address the following **Research Questions**:

1. What surface mineralization is naturally occurring in ultramafic mine tailings?
2. How is this mineralization impacted by the host mineralogy and grain-scale microstructure?
3. Does soil development (e.g., thickness and horizonation, organic carbon, clay formation) overcome passivating surface mineralization to enhance CDR?
4. What are low-cost, effective mitigation strategies to enhance CDR in natural mine tailings in temperate climates in the US and Canada?

We propose four sites of unstructured ultramafic mine tailings to answer these questions (Table 1). These sites were selected to evaluate the role of climate and mineralogy, with three sites having similar rainfall and temperatures, but different minerals (Quebec, Washington, Vermont), and one site as a climate outlier (California). In addition, our selected sites cover exposure timescales ranging from 150 years (Quebec, California) to present day, allowing us to quantify natural system mineralization rates and compare these results with lab-derived kinetic relationships.

Proposed Approach:

Table 1: Comparative Site Characteristics for Ultramafic Mine Tailings

Site location	Operational period	Primary commodity	Lithology	Serpentinization / mineralogy	Climate zone	Precipitation & temp
Thetford Mines, Quebec	1870s–2012	Asbestos (Chrysotile)	Dunite, harzburgite, pyroxenite, gabbro	**Variably serpentinized** Economic chrysotile in fractures; minor carbonates.	Humid Continental	1,171 mm/yr; -1.1 to 8.9 °C
Belvidere Mt, Vermont	1930s–1993	Asbestos (Chrysotile)	Dunite, harzburgite, serpentinite; in contact with metasediments	**Variably serpentinized**; Talc-carbonate & quartz-carbonate with magnesite/dolomite.	Humid Continental	1,370 mm/yr; 0.7 to 10.2 °C
Twin Sisters Mt, Washington	1950s–Present	Olivine	Unaltered dunite and harzburgite; cross-cutting pyroxenite veins	**Least serpentinized**; localized serpentinite in fault zones/landslides.	Oceanic - Alpine	1,550 mm/yr; 6.0 to 15.3 °C
New Idria, California	1854–1972	Mercury (Cinnabar)	Mantle diapir; contact between serpentinite and metasediments	**Fully serpentinized**; Chrysotile, lizardite, magnetite, and benitoite.	Semi-arid	385 mm/yr; 9.2 to 22.8 °C

Aim 1 'natural experiments': We will use natural mine tailing field sites (or quarries) to characterize the extent and rate of historic surface mineralization that has occurred over time and space across different rock types (mineral compositions, microstructures), exposure ages, climates and soils.

Methodology & Instrumentation: To bridge the gap between simple lab systems and more complicated natural systems, we will deploy a suite of high-resolution analytical techniques to characterize the "natural" samples from our four North American sites.

- **Mineralogical Fingerprinting:** We will use **X-Ray Diffraction (XRD)** to quantify bulk mineralogy and identify the presence of secondary carbonate phases or passivating clays.
- **Multi-scale Mapping:** To observe the passivation layers, we will couple **Micro-Computed Tomography (µCT**) with **petrography** to observe the internal fracture networks at the hand sample scale. **Scanning Electron Microscopy (SEM)** coupled with **Energy Dispersive Spectroscopy (EDS)** to map the thickness and elemental composition (Si, Fe, Mg) of coatings, and their correlation to microcrack and porosity structure, at the sub-micron scale.
- **Soil Development:** Soil profiles will be described, sampled, and evaluated in the lab for **grain size**, Fe-oxide and clay content using **XRD.** This will be coupled with **total organic carbon (TOC)** as a measure of soil respiration, CO_2 production, and carbonation.
- **Soil Carbon Flux:** In-situ soil development will be quantified using **soil probes and flux chambers** to measure pore-space chemistries (e.g., CO_2, pH, O_2, etc.) to relate microbial respiration to carbonate formation rates.

Aim 2 'controlled experiments': We will use representative materials from each site in Aim 1 to conduct laboratory experiments that test bulk mineralization reaction rates as a function of initial mineralogy and reactive surface area (grain size) under different temperature pressure, and CO_2 partial pressure conditions. The results of these experiments can then be directly correlated with reaction rates estimated by analyses from Aim 1.

Methodology & Instrumentation: Using the substrates characterized in Aim 1, we will isolate variables in a controlled setting to establish kinetic rate laws.

- **Batch Reaction Kinetics:** Substrates will be reacted in **high-pressure/temperature**

hydrothermal vessels. Fluids will be sampled at logarithmic intervals and analyzed via **Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES)** and **Carbonate Chemistry** to track cation release rates (SiO_2, Mg^{2+}, Ca^{2+}, Fe^{2+}), alkalinity and pH.
- **Mineral-Reaction Mapping: SEM, Transmission Electron Microscopy (TEM)** and **X-Ray Photoelectric Spectroscopy (XPS)** will be used to identify mineral passivation layers and thicknesses on the experimental products.
- **Geochemical Modeling:** All experimental data will be integrated into **Geochemist's Workbench (GWB)**. We will refine the standard kinetic databases using our site-specific Arrhenius parameters from the lab and 'field' experiments to create a predictive model for CO_2 uptake across our sites (and an inventory of other US and Canadian tailings locations).

Hypothesis Testing:

Aim 1: Natural Experiment Hypotheses

Determining what has already happened in the field over the last 150 years.

- **H1: The Depth-Age Decay Model.** We hypothesize that the extent of carbonate mineralization is a function of exposure age (time) and depth beneath the surface of the pile. Specifically, the highest mineralization concentrations will be at the surface of the oldest tailings, decaying exponentially as depth increases.
- **H2: Mineralogical Passivation Controls.** We hypothesize that the specific species of "secondary" minerals (passivation layers) are controlled by the host rock mineralogy:
    - **Olivine** and **orthopyroxene**, the main ultramafic silicates present in Thetford, Belvidere, and Twin Sisters, will produce passivating **Fe-oxyhydroxide** +/- **amorphous silica (SiO_2)** coatings.
    - **Serpentine, talc, +/- chlorite, brucite, carbonate,** the main serpentinization-related minerals present in Thetford, Belvidere, and New Idria, are more likely to bypass passivation to form stable **Ca- or Mg-carbonates**.
- **H3: Microstructural Penetration.** We hypothesize that "reaction-driven cracking" is influenced by the rock's initial grain-scale fracture density. Rocks with lower fracture strength (from coarser grain size, or lower strength phases from greater pre-existing alteration) will exhibit *higher* total carbonation and *deeper* penetration along micro-fractures.
- **H4: The Biogeochemical Catalyst.** We hypothesize that soil and regolith development *accelerate* mineralization through physical and chemical mechanisms.
    - Soils and plants will enhance moisture and promote permeability via root wedging, clay swelling, etc., naturally 'grinding' the rock piles.
    - Microbial and root respiration flux will increase pore-space pCO_2, enhancing subsurface rock carbonation and potentially limit passivating surface mineralization.
    - Seasonal retention of water and alternating redox environments in soil can flip Fe-oxides from passivating to enhancing weathering of ultramafic minerals as a function of natural Fe-humus complex development.

Aim 2: Controlled Experiment Hypotheses

Testing how we can "force" the system to perform better in the lab.

- **H5: The Temperature-Passivation Tradeoff.** We hypothesize that at ambient pCO_2 (~420 ppm), increasing temperature will increase reaction kinetics but primarily result in *thicker*

passivating layers (Si/Fe coatings), ultimately decreasing the total CO_2 storage potential over time.

- **H6: The pCO_2 Fugacity Threshold.** We hypothesize that there is a "tipping point" for pCO_2 fugacity (likely near 10,000 ppm) where the chemical drive to precipitate stable carbonates (magnesite/dolomite) overcomes the kinetic barrier of the passivation layer.
- **H7: Volume-Expansion Feedback.** We hypothesize that high-pCO_2 conditions will trigger a positive volume reaction (carbonates take up more space than silicates). This reaction-driven stress will generate new surface area via micro-cracking, enhancing the CDR reaction and inhibiting "clogging" of the reactive surface.

The "Null" Hypothesis

Providing a clear path regardless of the result.

- **H0:** We hypothesize that if mineralization remains insensitive to mineralogy, age, or climate across all four sites, it would prove that **environmental conditions are not the primary bottleneck**. This would support a "Universal Strategy" for CDR, suggesting that simple physical interventions (like turning the tailings) would be more effective than site-specific chemical management.

Scale and Impact

We estimate the exposed surface area and waste tonnage available at each site using AI (Table 2). Our four sites represent a diverse cross-section of the billions of tons of ultramafic tailings available in temperate regions globally. For example, the Thetford Mines region alone holds hundreds of millions of tons of reactive serpentinite. By defining the "tipping points" for practical carbonation of these specific mineralogies, our predictive framework can be exported to similar ophiolite complexes in Oman, Norway, and Australia, de-risking the implementation of passive surficial CDR at the gigaton scale.

Table 2: Estimated Tailing Footprints and Tonnages

Site Location	Estimated Surface Area (Exposed)	Estimated Waste Tonnage	Notes On Scale
Thetford Mines, Quebec	~2,000+ hectares (~5,000+ acres)	Hundreds of Millions of Tons	Extensive "blue-grey" hills dominate the landscape; multiple pits and vast debris fields across the region.
Belvidere Mt, Vermont	~623 hectares (~1,540 acres)	~29 Million Tons	Waste rock and mill tailings are scattered across the mountain; specifically includes two major steep tailings piles.
Twin Sisters Mt, Washington	~20–40 hectares (~50–100 acres)	Currently Unknown	Active mining site; footprint is smaller and more concentrated around the Millbank Material operation compared to the older asbestos giants.
New Idria, California	~16 hectares (~40 acres)	0.5 to 2 Million Tons	Concentrated "calcine" tailing piles (roasted cinnabar) and waste rock; secondary focus on mercury-contaminated sediment.

Differentiation: Why This Approach is Best-in-Class

While several organizations are currently exploring mine tailings for CDR, our project is uniquely

differentiated by its **natural-to-controlled experiment 'bridge'.** Most current efforts either focus on:

1. **Pure Lab Kinetics:** Which often fail when applied to heterogeneous field piles.
2. **Commercial Pilot Projects:** Which lack the granular, mineral-specific data to explain *why* carbonation rates slow down over time (passivation).

Our project is differentiated by three key factors:

1. **The "Time" Advantage:** Most CDR startups focus on "fresh" tailings. By using historic sites (e.g., Thetford, New Idria) as multi-decade "natural experiments," we can observe the long-term evolution of mineral coatings that are difficult or impossible to resolve in shorter pilot projects. Our approach moves beyond "if" carbonation happens to "how" it evolves over a 100-year horizon.
2. **Addressing the Passivation Bottleneck:** Instead of relying on energy-intensive mechanical grinding to expose new surfaces, our focus is on the **chemical and environmental triggers** of passivation. If we can optimize pCO_2 and moisture to bypass Si-coatings chemically, we reduce the energy penalty of CDR.
3. **Holistic System Integration:** We are not just looking at rocks; we are looking at the **lithosphere-biosphere-atmosphere interface**. By including soil/regolith development and microbial respiration in our model, we account for the natural "turbocharging" of carbonation that occurs in real-world environments.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include **scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks**. <500 words

The ability to make the measurements we propose is low-risk. Access to samples, personnel, and system complexity are the main risks.

Access to samples: We will need to gain access to mine tailings sites and permissions to collect rock and soil samples. Given Frontier's relationships and outreach, including development of activities at Thetford Mines, we anticipate that our approach could be applied to a diversity of sites as necessary. We anticipate gaining access to the olivine mine at Twin Sisters, which has been open to scientists for sampling in the past. However, if we are not granted access then we can easily pivot and use samples in our research collections (e.g., Twin Sisters Washington, Samail Ophiolite Oman, Josephine Ophiolite Oregon, Syros Greece) or to collect similar lithologies in non-mine tailing locations with public land access (possible for Belvidere, Twin Sisters, New Idria sites, and many others). We could target recently exposed surfaces (e.g., landslides, roadcuts) to control for mineralization over similar timescales to the mining operations.

Personnel: The work requires the timely recruitment of two postdoctoral researchers with the desired expertise to conduct the proposed research. We do not anticipate finding a postdoc risky, but their availability may fall outside of the target milestone dates. We would work with Frontier to renegotiate

a timeline if timing of postdoc recruitment becomes an issue.

System Complexity:

1. Scaling from lab to field conditions: Reaction kinetics measured under controlled conditions may not directly translate to heterogeneous natural systems. We will sample feedstocks in a manner where heterogeneity in initial composition (mineralogy, initial microstructure, and bulk geochemistry) and existing alteration/carbonation can be accounted for.

2. Environmental variability: Fluctuations in temperature, water availability, and geochemistry can produce nonlinear effects on mineralization rates. Our experiments are designed to explicitly test for these nuances across a suite of environmental conditions.

3. Quantification of reactive surface area in situ: Accurately measuring how surface area and passivation evolve during surficial mineralization and weathering reactions is a major uncertainty. By performing parallel experiments and utilizing microCT methods we can quantify changes in reactive surface area. Related characterization of associated soil will include identification and quantification of clay and iron-humus complexes. These soil-forming factors from ultramafic protoliths can increase soil organic carbon, soil respiration and CO_2, thereby enhancing subsurface weathering and carbonation.

While these are risks for our study, the outcomes of our work make important steps toward a systems-level understanding that can be built-upon and de-risked in future studies. Another way of looking at this issue is that the inherent risks of our study are one-and-the-same as the bottlenecks to advancing practical CDR - and thus provide the motivation to dig deeper into these problems.

e. **Proposal to Frontier:** Please list your **primary research deliverables**, **project timeline**, and **requested funding below**. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	We will produce a suite of novel datasets and integrated models to address uncertainties in CDR potential and engineering in unstructured mine tailings. **Datasets**: a) Identification of internal and surface mineralogy of mine tailings and related soils through depth profiles and different aged exposures to identify the nature and extent of carbonation reactions. b) Correlation matrix of mineralogy with quantifiable microstructural parameters (grain size, crack density, fraction of pre-existing alteration) to constrain the processes that limit the reaction extents and rates. c) Quantification of soil development (e.g., horizonation, clay formation, soil organic carbon accumulation), and CO_2 flux co-located with measurements of sub-surface mineralization and carbonation. d) Characterization of the chemical evolution, surface mineralization, and reaction rates at a range of temperatures and CO_2 partial pressure during

	the controlled experiments. **Models**: a) Compare and calibrate mineralization reaction rates for each site based on rock mineralogy, pressure, temperature, and CO_2 partial pressure across the 'natural' and controlled experiment timescales. b) Reactive transport model that calculates the onset and thickness of passivating mineral coatings over time. c) Quantify how soil development and microbial respiration accelerate subsurface mineralization. d) Optimize dual-strategy pathway for coordinated inorganic/organic mineralization. **Synthesis**: a) **CDR deployment decision tree & site evaluation toolkit:** A diagnostic framework that inputs site-specific mineralogy, climate data, and microstructure to output "reactivity readiness" scores. This approach bridges the gap between lab kinetics and field execution, providing the industry with a "go/no-go" logic for investing in surficial mineralization pathways. For example, this tool could identify when to use low-cost chemical interventions versus energy-intensive mechanical tilling, directly reducing the cost-per-ton of carbon sequestered when utilizing ultramafic deposits/mineral assemblages. b) **Two to three manuscripts / reports**. Example journals include: *ACS Environmental Science & Technology; Applied Geochemistry; Chemical Geology.* Topics would be: i) Kinetics and mitigation of mineral passivation and carbonate in ultramafic tailings; ii) Long-term carbonation rates in temperature ultramafic mine waste; iii) A scalable framework for evaluating surficial mineralization potential.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	The following are direct impacts for mineralization: 1) Knowing the surface mineralization in relation to mineralogy and grain-scale microstructures will enable targeted and optimized mitigation strategies. 2) Determining natural background weathering rates in ultramafic mine tailings over different exposure periods are key to inform interventions required for increased reaction rates. 3) Determining the development of soil weathering profiles and their role as a CO_2 sink can help engineer solutions to enhance CDR and mineralization. 4) Experimentally determined reaction rates, kinetic rate parameters and temperature dependent Arrhenius relationships will allow for accurate forward modeling of CDR and mineralization. Comparison

	of these relationships with our estimates of natural reaction rates from the mine settings will ground truth the application of lab data to CDR engineering problems. Together, these impacts aim to directly reduce measurement uncertainties and contribute to efficiency gains and scalability.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Milestone 1.1 Field work and sample collection; 7/1/2026 to 3/31/2027 Milestone 1.2 Characterize natural samples; 10/1/2026 to 7/31/2027 Milestone 1.3 Define key feedbacks in natural systems; 4/1/2027 to 12/31/2027 Milestone 2.1 Experimental design and testing; 1/1/2027 to 4/30/2027 Milestone 2.2 Run controlled experiments; 3/31/2027 to 12/31/2027 Milestone 1.4 / 2.3 Dissemination of results; 8/1/2027 to 12/31/2027 Milestone 2.4 Define key feedbacks in controlled experiments; 10/1/2027 to 4/30/2028 Milestone 1.5 / 2.5 Synthesize data from both studies; 9/1/2027 to 6/30/2028
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$745,681